

Software Requirements Specification

MedMind — medication reminders and medication status for the caregivers of older adults

Introduction to Software Engineering (15031001) · Milestone 2 — Requirements & Specification

Field	Value
Team name	MedMind
Project title	MedMind — medication reminders and medication status for the caregivers of older adults
SDLC chosen in M1 + reason	Incremental — build the core loop first (remind → confirm → alert the caregiver, FR-03–05), then add pairing and OCR (FR-01, FR-02).
Date / version	v2.0 · 10 October 2026
GitHub repo	https://github.com/yaroo393-png/SE-Project
Running demo	Not available yet — added when the prototype is ready

1. Introduction

1.1 Purpose and scope

Older adults who take medication regularly may forget a dose, take it at the wrong time, or not remember whether they have already taken it, while their children go out to work and cannot follow up all day. Current methods — alarm clocks, pill boxes, phone calls and LINE messages — cannot confirm the medication status and report it to the caregiver automatically.

MedMind answers exactly two questions: "**What medicine do I take at this meal, and have I taken it?**" (older adult) and "**Has every meal's medicine been taken today, without me having to call?**" (caregiver). The system links the older adult with the caregiver, helps record medicine details from the label with OCR while the caregiver checks them before saving, reminds the older adult at each medication meal, lets them confirm the whole meal with one tap, and alerts the caregiver when a meal is still not confirmed after three reminders. The MVP is five FRs. Success means the caregiver no longer has to call or send a LINE message for every meal, and knows when a meal has not been confirmed (success measures in Charter §4).

1.2 Users and stakeholders

Name / role	Kind	What they need	What they fear / forbid
Older adult (Senior)	primary user	A reminder sound, large text, large buttons, few taps; their child sets it up for them (interviewees 1, 3). Medicine photos: not in the MVP by team decision (FR-08, Could)	Having to sign up and enter data themselves (interviewee 3)
Child / caregiver	primary user	An alert only when a meal is not yet confirmed; photograph a label and have the details filled in (interviewee 2)	Having to check every meal; wrong medicine details from OCR
Pharmacist / hospital (label issuer)	secondary stakeholder (sets the rules)	Label instructions turned into reminder times exactly as written	The system interpreting, changing or recommending how a medicine is taken (§6)

Name / role	Kind	What they need	What they fear / forbid
External AI label-reading provider	external system	A clear label photo	An outage or unreadable label — manual entry must still work (FR-02, criterion 5); leaking medication data, which is health data (NFR-09)

2. Overall description

2.1 Product context — what it does / does not

MedMind links one older adult with one caregiver, turns caregiver-checked medicine details into medication meals, and tracks whether each meal is confirmed. The table below is the whole scope for this term; anything not listed is out.

IN SCOPE (MVP this term)	OUT OF SCOPE (explicit promise)
• FR-01 Pair older adult and caregiver (QR / invite code) • FR-02 Scan a label → draft → caregiver checks → save • FR-03 Reminder at each meal + repeat reminders • FR-04 Confirm the whole meal • FR-05 Caregiver alert + daily status	• OCR of doctors' handwriting • Saving OCR results without a check • AI diagnosis, medicine advice, or advice on making up a missed dose • The system adding, reducing or changing how a medicine is taken • Post-MVP features (Should / Could): see §3.1 and §3.7

2.2 Assumptions and constraints

- Constraint: confirming is self-reported; it is not proof that the medicine was taken.
- Constraint: every OCR result must be checked by the caregiver before it is saved.
- Constraint: the MVP supports only tablets and capsules taken at fixed times — not as-needed medicines, liquids or half tablets.
- Constraint: the M3 prototype is a clickable web page with no real backend; reminders, time and OCR are simulated.
- Constraint: the system is a web app used through a browser. In the MVP the older adult is reminded and the caregiver is alerted **only while the web page is open**; when the page is closed or the screen is locked, no alert is given (not in the MVP).
- Constraint: browsers do not play sound until the user has tapped the page at least once after opening it, so the reminder sound plays only after that first tap.
- Constraint: scanning a QR code or photographing a label needs camera permission; without it, the 6-digit invite code or manual entry (FR-02, criterion 5) is used instead.
- Assumption: the older adult and the caregiver have smartphones that can open the web app and keep it open during medication times — not yet verified with users.
- Assumption: the caregiver does all of the main setup (interviewee 3).
- Constraint: in the MVP one older adult has one caregiver (several caregivers is a Should, after the MVP).
- Constraint: reminder times come from the label instructions only; the system never guesses a time.

3. Functional requirements

Format: As a <role>, I want <goal>, so that <benefit>. Every FR traces back to the problem and the interviews, and every acceptance criterion covers either the correct path or a wrong / edge path, written Given–When–Then so that it can be checked by a test.

3.1 Priority summary (MoSCoW)

ID	Requirement	Priority	Why this priority
FR-01	Pair older adult and caregiver by QR / invite code	Must	Answers question 2
FR-02	Scan a medication label and create the medicine record	Must	Answers question 1
FR-03	Reminder at each meal + repeat reminders	Must	Answers question 1 — the heart of the MVP
FR-04	Confirm the whole meal with one tap	Must	Answers questions 1 and 2
FR-05	Caregiver alert + daily status	Must	Answers question 2
FR-06	Several caregivers per older adult	Should	After the MVP — the MVP supports one caregiver
FR-07	List of all of the older adult's medicines	Could	The older adult already sees the meal's medicines on the reminder screen (FR-03); this page is optional
FR-08	Medicine photos on the reminder screen	Could	Interviewee 1 asked for photos — for the MVP the team chose a reminder sound only

Five Must requirements (the course sets the MVP at 5 FRs), chosen so that both questions in §1.1 are answered end to end. FR-06 to FR-08 are not Must: each one can wait until after the MVP and the system still answers both questions. Won't: see §6.

3.2 FR-01 — Pair older adult and caregiver [Must]

User story: *As a caregiver, I want to pair with my parent's phone by scanning a QR code or entering an invite code, so that I can set up medicines and receive their status without my parent creating an account.*

Pain this traces to: the caregiver is not there and has to call; older adults find it hard to sign up for an account (interviewee 3).
Definition: the invite code (QR + 6 digits) works once and expires after 10 minutes; the older adult can tap "Create new code". The older adult must tap consent before pairing and has no account or password. In the MVP an older adult has one caregiver. Either side can unpair (with a second confirmation); the caregiver then loses access immediately and that pair's medicines and confirmation history are deleted (PDPA right to erasure).

1. **Given** the older adult opens the app for the first time, **when** the start screen appears, **then** it shows a QR code and an invite code without any sign-up or typing.
2. **Wrong path. Given** the older adult has not tapped consent, **when** a caregiver tries to pair, **then** pairing fails and the caregiver is told it is waiting for consent.
3. **Given** the older adult has consented, **when** the caregiver scans or enters a valid code, **then** both screens show that they are connected.
4. **Wrong path. Given** a code that is wrong, older than 10 minutes, or already used, **when** the caregiver enters it, **then** it is rejected with a clear message and no connection is made.
5. **Wrong path. Given** the older adult is already paired with a caregiver, **when** a second caregiver scans or enters the code, **then** it is rejected with "This older adult already has a caregiver" and no connection is made.
6. **Wrong path. Given** a paired older adult and caregiver, **when** either of them unpairs and confirms, **then** the caregiver immediately can no longer see the older adult's data, the pair's medicines and confirmation history are deleted, and the older adult sees a new QR code.

3.3 FR-02 — Scan a medication label and create the medicine record [Must]

User story: *As a caregiver, I want to photograph a medication label and get a pre-filled draft that I can check and correct, so that I don't have to type every medicine by hand while staying in control of what is saved.*

Pain this traces to: the caregiver wants to photograph the label and have it filled in (interviewee 2); the older adult wants their child to set it up (interviewee 3).

Definition: only the caregiver can add or edit medicines. Required fields are medicine name, tablets per dose, and meal / time; warnings and contraindications are saved only when the label has them. The reminder time follows the label only: after a meal = the meal time; N minutes before a meal = the meal time – N; at bedtime = bedtime. The caregiver sets breakfast, lunch and dinner times and bedtime when adding the first medicine. Medicines with the same reminder time are grouped into one meal.

1. **Given** a clear label photo, **when** the caregiver submits it, **then** a draft is shown (medicine name, tablets per dose, meal / time, warnings, contraindications) and nothing is saved yet.
2. **Given** the caregiver edits a field in the draft, **when** they tap save, **then** the edited value is saved, not the value read from the photo.
3. **Wrong path. Given** a field cannot be read, **when** the draft is shown, **then** that field is empty and marked "Please check" (never guessed), and saving is blocked until every required field is filled.
4. **Wrong path. Given** a label for a medicine taken only when needed, **when** the draft is shown, **then** the system says "As-needed medicines are not supported yet" and creates no schedule.
5. **Wrong path. Given** the scan fails, **when** the system cannot read the photo, **then** an empty form opens for manual entry.
6. **Given** the first medicine is added with the label "after breakfast", **when** the caregiver sets the breakfast, lunch and dinner times and bedtime (breakfast 07:00), **then** the reminder is at 07:00.
7. **Wrong path. Given** the caregiver is on the draft screen, **when** they tap cancel, **then** nothing is saved.
8. **Given** the label "30 minutes before breakfast" and breakfast at 07:00, **when** it is saved, **then** the reminder is at 06:30.
9. **Given** the label "at bedtime" and bedtime at 21:00, **when** it is saved, **then** the reminder is at 21:00.
10. **Wrong path. Given** the label says "before meals" without a number of minutes, **when** the draft is shown, **then** the minutes field is empty and marked "Please check" (the system does not guess), and saving is blocked until it is filled.

3.4 FR-03 — Reminder at each meal [Must]

User story: As an older adult, I want a reminder at each meal time that shows all of that meal's medicines on one screen, and to be reminded again if I haven't confirmed, so that I don't forget or mix up my medicines.

Pain this traces to: forgetting a dose or taking it late (problem statement); switching the alarm off and then forgetting (interviewee 1).

Definition — reminder timeline, shared with FR-04 and FR-05 (T = the meal's reminder time):

T	first reminder	[Taken] [Not yet]
T+5	repeat reminder 1	(if not confirmed)
T+10	repeat reminder 2	(last round; "Not yet" = alert the caregiver now)
T+15	not confirmed -> caregiver alerted "Medicine not yet confirmed" (unless already alerted at T+10)	
After T+15	-> late confirmation allowed until the reminder time of the next meal that starts after T+15 (none = until 23:59) -> caregiver told "Medicine confirmed"	

Repeat reminders are at fixed times from T and only while the meal is not confirmed. The reminder screen shows every medicine in the meal (name, tablets, warnings, contraindications), a green "Taken" and a red "Not yet" button, and plays a sound at every round while MedMind is open; it shows no medicine photos. The caregiver alert is sent once per meal and never says "not taken". Meals that overlap (e.g. 06:45 before breakfast and 07:00 after breakfast) each keep their own reminders, caregiver alert and late-confirmation window; the screen lists every unconfirmed meal in time order under separate headings, each with its own buttons, and reminders due at the same moment sound once.

1. **Given** an 08:00 meal with two medicines and MedMind open on the older adult's phone, **when** it is 08:00, **then** a reminder sound plays and both medicines (name, tablets, warnings, contraindications) are shown on one screen with "Taken" / "Not yet" buttons and no medicine photos.

2. **Wrong path. Given** the meal is not confirmed, **when** it is 08:05 and then 08:10, **then** repeat reminders 1 and 2 are given, each with sound.
3. **Given** the meal is confirmed, **when** the next repeat-reminder time arrives, **then** no reminder is given.
4. **Wrong path. Given** meal A at 06:45 (15 minutes before breakfast) is not confirmed and meal B is at 07:00, **when** it is 07:00, **then** the reminder screen shows both meals in time order under separate headings with separate buttons and one sound, and the caregiver is alerted about meal A at 07:00, on meal A's own timeline.

3.5 FR-04 — Confirm the whole meal [Must]

User story: *As an older adult, I want to confirm the whole meal with one tap, so that I don't have to confirm each pill and my family knows I reported taking them.*

Pain this traces to: not remembering whether the medicine was taken (problem statement); wanting an easy way to tell their child (interviewee 1); large buttons and few taps (interviewee 3).

Definition: the older adult confirms on the reminder screen (FR-03) or on their home screen, which shows today's meals with their status and a "Taken" button for any meal still in the late-confirmation window.

1. **Given** the 08:00 meal reminder is showing, **when** the older adult taps "Taken" at 08:03, **then** the status becomes "Taken", 08:03 is recorded, and reminders for that meal stop.
2. **Wrong path. Given** the meal is already confirmed, **when** "Taken" is tapped again, **then** no duplicate record is created.
3. **Given** the first reminder at 08:00, **when** the older adult taps "Not yet", **then** they are reminded again at 08:05.
4. **Wrong path. Given** the last reminder at 08:10, **when** the older adult taps "Not yet", **then** the caregiver is alerted immediately (without waiting for 08:15).
5. **Given** the meal was not confirmed by 08:15, **when** the older adult taps "Taken" on the home screen at 08:40, **then** the status becomes "Taken (late confirmation)" with 08:40.
6. **Wrong path. Given** the next meal's reminder time (12:00) has passed, **when** the older adult tries to confirm the 08:00 meal, **then** it cannot be confirmed and the status stays "Not yet confirmed".

3.6 FR-05 — Caregiver alert and daily status [Must]

User story: *As a caregiver, I want to be notified only when a meal is still not confirmed after the reminders, and to see today's status of every meal in one place, so that I don't have to call or message for every meal.*

Pain this traces to: having to call or send a LINE message for every meal; no single place showing the status (problem statement); wanting an alert only when a meal is not yet confirmed (interviewee 2).

1. **Given** the older adult confirmed at 08:12, **when** it is 08:15, **then** the caregiver is not alerted.
2. **Wrong path. Given** there is no confirmation, **when** it is 08:15, **then** the caregiver is alerted "Medicine not yet confirmed" (never "not taken").
3. **Given** several meals in the day, **when** the caregiver opens the status page, **then** every meal of the day is shown as "Taken" + time, "Taken (late confirmation)" + time, "Not yet confirmed", or "Not due yet".
4. **Wrong path. Given** there are no medication meals today, **when** the caregiver opens the status page, **then** it shows "No medication meals today".
5. **Given** the older adult tapped "Not yet" in the 08:10 round (the caregiver has already been alerted), **when** it is 08:15, **then** the caregiver is not alerted again.
6. **Given** the caregiver has been alerted "Medicine not yet confirmed", **when** the older adult confirms late at 08:40, **then** the caregiver receives "Medicine confirmed at 08:40".

3.7 Should and Could requirements

These are specified at story level for this milestone; full acceptance criteria are written when each one is moved into an increment.

ID	Priority	User story
FR-06	Should	As a caregiver who shares care with my siblings, I want more than one caregiver to pair with the same older adult, so that whoever is free at that time can see the status and follow up.
FR-07	Could	As an older adult, I want to see all my medicines and their times for the day, so that I know what I take today without waiting for a reminder.
FR-08	Could	As an older adult, I want to see a photo of each medicine on the reminder, so that I pick the right pills.

4. Non-functional requirements

Forbidden words unless quantified: fast, easy, user-friendly, smart, secure. NFR-04 is the required security / privacy NFR. The numbers are the team's targets (8 October 2026).

ID	Kind	Statement (testable)	How we will check
NFR-01	reliability	Repeat reminders occur at T+5 and T+10, and the caregiver is alerted no later than T+15 every time a meal is not confirmed (at T+15, or immediately when "Not yet" is tapped in the T+10 round), only once per meal.	Automated test with a simulated clock
NFR-02	usability	At least 4 of 5 older adults confirm within 10 seconds of seeing the reminder screen, on their first try and without help (if only 3 can be found: 3 of 3, reported as a small sample).	Hand over the prototype on the reminder screen, say only "If you have taken your medicine, what do you do next?", and time it — when prototype Increment 1 is done
NFR-03	usability	"Taken" / "Not yet" buttons at least 64 px high and full width · medicine names on the reminder screen at least 24 px · body text at least 18 px · text-to-background contrast at least 4.5:1 (standard minimum button size is 44–48 px in WCAG 2.2 AAA / Apple / Material — larger here for older adults).	Measure in the browser's DevTools and a contrast checker — when prototype Increment 1 is done
NFR-04	security / privacy (required)	A caregiver sees only the older adult they are paired with; a request for another older adult's data is refused.	Automated test: change the older adult's ID and confirm the request is refused (in the logic module)
NFR-05	privacy	No connection happens before the older adult taps consent (0 times).	Automated test (FR-01, criterion 2)
NFR-06	safety	No medicine record is saved without the caregiver's check (0 records).	Automated test (FR-02)
NFR-07	reliability	However many times "Taken" is tapped, there is only one confirmation record for that meal.	Automated test (FR-04, criterion 2)
NFR-08	performance	The draft read from a label photo appears within 10 seconds of submitting the photo for at least 18 of 20 labels (90%).	Time from submitting the photo to seeing the draft, with 20 real labels — not this term: the prototype simulates label reading, so it is checked once real label reading exists (Should)

ID	Kind	Statement (testable)	How we will check
NFR-09	privacy / PDPA	The system stores only what FR-01–05 use (medicine details, meal times, confirmation status, pairing) and stores none of the older adult's national ID number, address, date of birth, phone number or hospital number (0 fields).	Review every field in the data model (Design — architecture.md §5), when the data model is done
NFR-10	privacy / PDPA	After unpairing, the former caregiver's requests for that data succeed 0 times , and that pair's medicines and history are 0 records .	Automated test: after unpairing, the request must be refused and the remaining records counted

5. Use cases

5.1 Use-case list

Use case	Actor	Related FR	Brief success story
UC-1 Pair devices	Caregiver, Older adult	FR-01	The older adult taps consent → the caregiver scans the QR → both screens show "connected".
UC-2 Add medicine from a label	Caregiver	FR-02	The caregiver photographs a label, fixes one field in the draft and taps save → the medicine and its meal are saved.
UC-3 Receive a meal reminder	System → Older adult	FR-03	At 08:00 the older adult sees both morning medicines on one screen.
UC-4 Confirm medicine taken	Older adult	FR-04	The older adult taps "Taken" → the meal is confirmed and reminders stop.
UC-5 Alert the caregiver when not confirmed	System → Caregiver	FR-05	No confirmation by T+15 → the caregiver receives "Medicine not yet confirmed".
UC-6 View daily status	Caregiver	FR-05	The caregiver opens the status page and sees every meal of the day with its time.

One use case per Must requirement; FR-05 has two (UC-5 and UC-6) because it combines the caregiver alert and the daily status.

5.2 Use-case diagram

Not drawn in this document. The use-case diagram belongs to the Design document (docs/03-design/architecture.md) and has not been drawn yet.

6. Out of scope

1. OCR of doctors' handwriting — easily misread, giving wrong medicine data, so the older adult could take the wrong medicine.
2. Saving OCR results without a check — wrong medicine data would enter the system unchecked; the caregiver must check every time.
3. AI diagnosis or medicine advice — the team are not medical professionals, and wrong advice could harm the patient.
4. The system adding, reducing or changing how a medicine is taken — it must follow the label or the doctor's order only.
5. Advice on making up a missed dose — e.g. taking a double dose could be dangerous; it is medical advice.

7. Traceability matrix (Golden Thread)

The FR-03 row (the heart of the MVP) is filled across every column as *planned*, following the instructor's example. FR-06–08 are outside the MVP (§3.7). The other rows are filled when the Design (architecture.md) and the prototype are done, and the Test column becomes real results once the tests have run.

Problem / pain	Requirement	Solution / design	Feature built	Test
Older adults find sign-up hard; the caregiver is not there	FR-01	(Design — architecture.md, pending)	(M3 prototype)	FR-01 criteria 1–6 · NFR-04, NFR-05, NFR-10
The caregiver wants help filling in medicine details from the label	FR-02	(Design — architecture.md, pending)	(M3 prototype)	FR-02 criteria 1–10 · NFR-06, NFR-09
Forgetting a dose / taking it late	FR-03	State machine of a meal: waiting → reminding (T, T+5, T+10) → taken / not yet confirmed / late confirmation (<i>planned</i>)	Reminder screen: every medicine in the meal + sound + "Taken" / "Not yet" buttons (<i>planned</i>)	FR-03 criteria 1–4 · NFR-01 (<i>planned</i>)
Not remembering whether the medicine was taken	FR-04	(Design — architecture.md, pending)	(M3 prototype)	FR-04 criteria 1–6 · NFR-02, NFR-03, NFR-07
Calling or messaging for every meal; no single status	FR-05	(Design — architecture.md, pending)	(M3 prototype)	FR-05 criteria 1–6 · NFR-01
The caregiver is not always around and sometimes cannot call	FR-06 (Should)	(after the MVP)	(after the MVP)	(after the MVP — criteria written when moved into an increment)
Not knowing which medicines to take at this meal / today	FR-07 (Could)	(after the MVP)	(after the MVP)	(after the MVP — criteria written when moved into an increment)
Wanting to see medicine photos (interviewee 1)	FR-08 (Could)	(after the MVP)	(after the MVP)	(after the MVP — criteria written when moved into an increment)

8. AI usage log

Every time AI drafted content it is logged honestly (if this table is empty but the content looks AI-written, the milestone is capped at 50%, per the instructor's SRS template). This table shows the key entries where AI drafted or changed the content of this SRS; the full log is in the repository (docs/02-requirements/spec.md §8).

Date	Tool / model	What we asked	What AI produced	What we changed / verified	What we learned
7–8 Oct 2026	Claude (Claude Code)	Draft the user stories and acceptance criteria for our FRs, together with the Charter and this SRS, from our team's new problem statement.	A first Charter with six FRs, including two that are not in our problem statement: medicine stock with low-stock alerts (tied to a pain no interview had confirmed) and viewing / editing the medication history. It also filled in choices we had not made (a "Not yet" button, meal times) and later carried content over from our old M1.	Cut to the five FRs of our problem statement; stock and history editing became Should items after the MVP. Had the old M1 content reverted and rebuilt the documents from the new problem statement only. Decided the reminder rules, the buttons and the caregiver messages ourselves.	AI pads the scope and reaches for old material. Every FR must trace to a pain we actually heard in an interview, and the team decides the scope and the numbers.
8 Oct 2026	Claude (Claude Code)	Suggest numbers for NFR-02/03/08 and success measure 6.	Values with sources (WCAG 2.2 / Apple / Material for button size, 5 users following Nielsen) and a proposal to split measure 6 into this term / after the MVP.	The team asked "how do we test with a prototype that cannot be clicked yet?" → stated that these are measured when prototype Increment 1 is done; the baseline for measure 6 had to come from interviewee 2 (AI does not guess).	A number must say when and how it is measured, or it looks as if it has already been measured.
8 Oct 2026	Claude (Claude Code)	Check the SRS against the M2 guide and the example SRS (.docx).	Found only 2 stakeholders (needs at least 3), no priority in the SRS, and PDPA covering consent only. AI had wrongly deleted the "capped at 50%" sentence from the instructor's template, and proposed "no screens in FR-04", then withdrew it after re-checking the instructor's example.	The team asked to add 2 stakeholders, a priority on every FR plus a MoSCoW table, NFR-09/10 with the rule that unpairing deletes the data (FR-01, criterion 6), and to restore the 50% sentence. The .docx was our old MedMind version, used for its shape only, not its content.	Check against the instructor's source documents before removing or adding a rule, and read the examples every time.
8 Oct 2026	Claude (Claude Code)	Re-check the SRS against the M2 guide and the Final Project file (M4).	Every FR was Must (the guide says this means prioritising was never done), FR-05 combined two behaviours, and the Golden Thread table was empty; the two instructor documents disagree on M2 points and on where diagrams go.	The team asked to write FR-06 (Should) and FR-07–08 (Could) as user stories from the interviews, justify FR-05 in the quality check, and fill the FR-03 row as planned; the mismatch between documents will be asked to the instructor.	Priority must be visible on the FR itself, not only in the backlog.

Date	Tool / model	What we asked	What AI produced	What we changed / verified	What we learned
9 Oct 2026	Claude (Claude Code)	The project is a web app — what in the documents must change?	The documents still assumed the older adult "allows notifications" on a smartphone, but a web page cannot alert once it is closed or the screen is locked (that needs Web Push), browsers do not play sound before the first tap, and the camera (QR / label) needs permission.	The team kept a clickable web prototype with no server and chose an MVP that reminds only while the page is open → updated §2.2, the reminder-screen rule and FR-03 criterion 1; the number of criteria did not change.	Platform limits must be written into the assumptions and constraints before writing the criteria that depend on them.

9. Self-check

Scorecard item	Score	Evidence
1. Scope (in / out) is clear and traces back to the M1 problem	2	§2.1 and §6; the two questions in §1.1 come from the Charter
2. All actors that use the system are named	2	§1.2 — four roles, including the label issuer and the external AI provider
3. At least 5 FRs, numbered, written as user stories passing the 4 quality gates	2	§3 — 8 FRs, the 5 Must written in full in §3.2–§3.6
4. At least 3 NFRs, each with a number	2	§4 — 10 NFRs, each with a number or a checkable condition
5. Given–When–Then for every Must, including a wrong path	2	§3.2–§3.6 — 32 criteria, 16 of them wrong paths
6. MoSCoW applied, with 3–5 Must	2	§3.1 — exactly 5 Must, reason given per row
7. No solution or technology mixed into a requirement	2	No product or vendor named in an FR; platform limits kept in §2.2
8. Traceability: story ↔ FR ↔ acceptance	2	§7 — every FR traces to a pain; the FR-03 row is filled as planned
9. No gold-plating; every FR ties to a stated problem	2	"Pain this traces to" under every Must FR
10. Every FR passes the 4 quality gates (no vague words, no solution)	2	Named roles, "so that" clauses and numeric criteria throughout
Total	20 / 20	Target is at least 14 / 20

Course rule: you may use AI to draft. You must log it, verify it against a real stakeholder, and be able to explain every FR in the oral exam. Work you cannot explain earns no credit.